

Just the Right Amount of Exercise: Aerobic Exercise and Memory

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ABSTRACT

A popular perception is that aerobic exercises are effective for improving performance on memory tasks. However, previous studies have not yet explored the effectiveness of aerobic exercise for individuals. We investigate the effects of aerobic exercise on English vocabulary learning by managing the conditions under which aerobic exercise is effective for individuals. We use a heart-rate sensor to control the amount of exertion a participant made on a treadmill while learning the vocabulary words. The results show this method can improve memory retention. This indicates that close monitoring of individual aerobic exertion through physiological sensing can make improved learning with exercise a reality.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

augmented learning, ubiquitous sensing, physiological sensing

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1 INTRODUCTION

It is been posited that learning during and after aerobic exercise is effective in improving memory [3], and [2] showed that the ability to memorize words increased significantly after three months of exercise. This was attributed to the fact that aerobic exercise increased the blood flow to the hippocampus, an organ related to memory [5]. It is thought that the increased blood flow to the hippocampus increases the production of Brain Derived Neurotrophic Factor (BDNF), a protein that stimulates and maintains the growth

of nerve cells in the brain, thereby enhancing memory [3]. However, it is difficult to generalize the hypothesis for all individuals.

We investigate the relationship between aerobic exercise and memory with 24 participants who memorized English words while using a treadmill. The results showed that the percentage of correct answers improved by 7 percent after one day and by 9 percent after three days, and this difference was statistically significant. We also employed machine learning to estimate for whom aerobic exercise would be effective. Both recall and precision were 83.3%, successfully identifying such people.

2 ESTIMATING AEROBIC CAPABILITIES

In order to control aerobic exertion, we need to be able to measure users. We use the Heart Rate Reserve (HRR) [4], a method to calculate the appropriate heart rate to set the intensity of exercise. The HRR is the difference between the maximum heart rate and the resting heart rate, and is expressed by the equations $R = H_m - H_r$ and $H_m = 220 - a$ where R is the HRR, H_m and H_r are the maximum heart rate and resting heart rate, respectively, and a is the age. Using these equations, the target heart rate is expressed by the equation $T_H = [(H_m - H_r) \times I] + H_r$. Here, T_H is the target heart rate (Target HR) and I is the intensity as a percentage of the HRR.

The target minimum and maximum heart rates are set by adjusting I , and the exercise intensity is adjusted to keep the heart rate within this range.

In addition, factors other than the heart rate during exercise may be related to the effect of aerobic exercise. The Global Physical Activity Questionnaire (GPAQ) [1] is a questionnaire to evaluate the frequency and intensity of exercise per week.

3 EXPERIMENT

The participants were 24 Japanese undergraduate and graduate students (M=22, F=2), aged from 20 to 25 years old. These participants were asked to memorize English words with and without exercise. The experiment received ethical approval in advance.

3.1 Experimental Methods

We randomly divided the participants into Group A and Group B. We used a treadmill as shown in Figure 1 which allows us to close adjust the speed and gradient for each participant, and was adjusted with HRR. The target heart rate is set at a light exercise intensity I of $0.3 \leq I \leq 0.4$. Participants in Group A used the running machine and participants in Group B sat on a chair. Their

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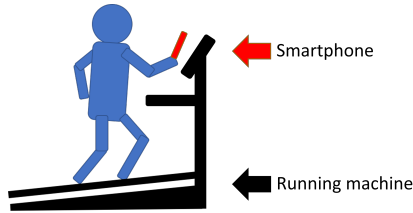
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Table 1: Mean and Difference of Correct Answer Rate and Result of Paired t-Test

	Correct Answers Avg. (Exercise)	Correct Answers Avg. (Sitting)	Difference	t-score	p-value差
After 5 Minutes	0.87	0.86	0.01	0.28	$p = 0.78$ でなし
After 1 Day	0.83	0.76	0.07	2.4	$p = 0.02$ であり
After 3 Days	0.70	0.60	0.10	3.4	$p = 0.003$ であり
After 7 Days	0.47	0.38	0.09	1.9	$p = 0.067$ でなし

**Figure 1: Experiment in Memorization using the Treadmill****Figure 2: Heart rate sensor used:WHS-3**

heart rates were recorded using the WHS-3 heart rate sensor¹ as shown in Figure 2. This sensor samples the heartbeat wave form at 128 Hz and heartbeat cycle at 1000 Hz.

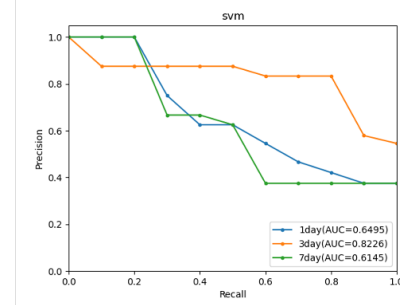
A flashcard application was used for the participants to memorize 60 English words each. Participants labeled the words they felt they had memorized as "correct" and unmemorized as "incorrect". The participants were also asked to answer the GPAQ. In the initial recall test, all 60 words were tested five minutes after the activity. After this, the 60 word sets were divided into three 20 word groups which were tested one group at a time after one, three, and seven days. Participants repeated the experiment using the opposite condition.

3.2 Results

Table 1 shows the results of paired t-tests and the mean values of the percentage of correct answers to the word test when learning during exercise and when learning while sitting. There is a significant difference 1 day and 3 days after memorization. We found that the aerobic exercise was effective with controlled HRR.

Next we explored the effectiveness of the aerobic exercise using a currently obtained HRR. If this is possible, we can selectively

recommend aerobic exercise to individuals. We created and evaluated a machine learning model that classifies participants. As features, we used the heart rate at rest, and the results from the GPAQ questionnaire. Using these features, SVM and random forest was for data collected after 1 day, 3 days, and 7 days, respectively. For SVM, both precision and recall were 83.3%, which is higher than that of random forest. The precision-recall curves for SVM are shown in Figure 3. The results show Day 1 and Day 7 do not have clear classification, but the results for Day 3 indicates that we can recommend for whom aerobic exercise is effective.

**Figure 3: Precision-Recall Curves for Each Day of Testing.**

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¹<https://www.uniontool-mybeat.com/SHOP/8600085.html>